

Teaching That Sticks: An Example of A Bad Slide (please do not do this)

An Investigation Into Sweller's Cognitive Load *Theory of Multimedia Learning* (1988) And Its Practical Application In Modern Pedagogy and *Web Content Accessibility Guidelines* (WCAG 2.1) Compliance Standards

Cognitive Load Theory (CLT) is an educational theory proposed by John Sweller in 1988. It posits that because human working memory has a limited capacity, instructional materials should be designed in a way that does not overload it. There are three types of cognitive load:

1. Intrinsic cognitive load, which is the effort associated with a specific topic
2. Extraneous cognitive load, which refers to the way information or tasks are presented to a learner; and
3. Germane cognitive load, which is the work put into creating a permanent store of knowledge, or a 'schema.'

When designing slides, one must ensure that extraneous load is minimized, intrinsic load is managed, and germane load is promoted. Furthermore, accessibility is critical. WCAG guidelines suggest contrast ratios should be at least 4.5:1 for normal text.



Further Reading and References:

- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.
- Mayer, R. E., & Moreno, R. (2003). Nine ways to reduce cognitive load in multimedia learning. *Educational Psychologist*, 38(1), 43–52.
- Paas, F., Renkl, A., & Sweller, J. (2003). Cognitive load theory and instructional design: Recent developments. *Educational Psychologist*, 38(1), 1–4.
- van Merriënboer, J. J. G., & Sweller, J. (2005). Cognitive load theory and complex learning: Recent developments and future directions. *Educational Psychology Review*, 17(2), 147–177.